Two barriers for minimal-time firing squads, with applications to four and five states

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Abstract

A minimal-time solution of the firing squad synchronization problem synchronizes every line of n cells at time $2n-2$. Six states suffice; whether five do has been open since 1987. Every line follows the infinite half-line until the echo of its right end returns. For any number of states, we prove that infinitely many rows parallel to the half-line's front become periodic only after crossing the line $i = t/3$, up to an arbitrarily small angle, and that the half-line is never periodic in time and space on a wedge at its left border, hence never eventually regular. With certified satisfiability checks, we exclude four states using the lines of lengths 2 to 5 and parts of those of lengths 6 to 10, give five-state rules synchronizing exactly the lengths 2 to 14, and show that every five-state solution uses at least 15 non-quiescent neighbourhoods on the first 78 anti-diagonals of its half-line.

Keywords: firing squad synchronization problem; cellular automata; minimal time; state complexity; lower bounds; pumping lemma; eventual periodicity; SAT solving; proof certificates

1 INTRODUCTION

The firing squad synchronization problem, posed by Myhill in 1957 and made popular by Moore [1], asks for a finite automaton with the following property. Copies of the automaton are placed in the cells of a line of arbitrary length n. At time 0 the leftmost cell, the *general*, is in a special state and all other cells are *quiescent*. At each step every cell updates its state from its own state and the states of its two neighbours. All cells must then enter a *firing* state at the same time, and none of them earlier. The difficulty is that the cells are identical, that each sees only its two neighbours, and that a cell with k states cannot count up to n: the same finite rule has to synchronize lines of every length.

No line can be synchronized before time $2n-2$: the order of the general needs $n-1$ steps to reach the far end of the line, and the news that the far end exists needs $n-1$ more steps to come back. Goto [2] and Waksman [3] showed that time $2n-2$ can be achieved, and a solution that synchronizes every line of length $n \geq 2$ at time $2n-2$ is called a *minimal-time solution*. Since then, the least number of states of a minimal-time solution has been a benchmark question of cellular automata theory [4, 5, 6]. Waksman's solution has 16 states, Balzer's 8 [7], Gerken's 7 [8] and Mazoyer's 6 [9]. In the other direction, Balzer [7] reported a computer search showing that no four-state minimal-time solution exists; Yunès [10] repeated the search and observed that the lines of lengths $2, \ldots, 9$ suffice, and Sanders [11] found Balzer's search incomplete and confirmed the result with a corrected one. Many six-state minimal-time solutions are known today [12, 13]; solutions with at most five states are known for restricted families of lengths, such as the powers of two, some of them in non-minimal time [14, 15], and four-state protocols synchronize the rings whose lengths are powers of two [16]. Whether a five-state minimal-time solution exists has been open since Mazoyer's construction [5, 6].

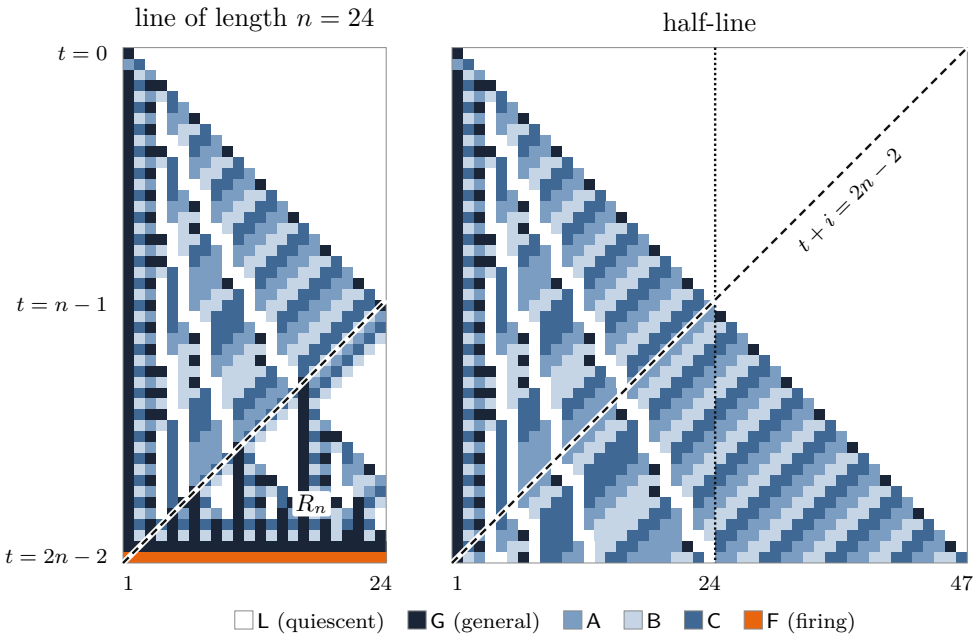


Figure 1: Mazoyer's six-state minimal-time solution. Left: the line of length $n = 24$; time runs downwards from $t = 0$ to the firing time $t = 2n - 2 = 46$, and the cells $i = 1, \ldots, 24$ run from left to right. Right: the half-line, which is started in the same way but has no right end, for the same times. The two diagrams coincide above the dashed anti-diagonal $t + i = 2n - 2$ (Lemma 2.2); the dotted line marks the right end of the line of length 24. Below the anti-diagonal lies the reflected triangle R_n, in which the line reacts to its right end; on the first anti-diagonal of R_n, the *return chain* $t + i = 2n - 1$, every cell of the line differs from the corresponding cell of the half-line (Lemma 2.4).

Why lower bounds are hard A five-state rule is a table of 96 entries (Section 2), two of which are fixed by quiescence, so there are $5^{94} > 10^{65}$ rules. A rule can be refuted by a single line that it fails to synchronize, but a minimal-time solution has to synchronize every line, so a proof that five states do not suffice must exclude all rules. As there are finitely many rules, it suffices to find one length N such that no five-state rule synchronizes the lines of lengths $2, \ldots, N$; a negative answer can thus consist of finitely many conditions. The difficulty is to find conditions that are strong enough and cheap enough to check. This paper proves such conditions for every number of states and measures how far they reach for four and five states.

The idea Until the echo of its right end comes back, a line cannot tell that it is finite: it follows the infinite *half-line*, which is started in the same way but has no right end (Lemma 2.2). Figure 1 shows this for Mazoyer's solution. A minimal-time solution therefore consists of a single space-time diagram of the half-line, shared by all lengths, and, for each length n, a *reflected triangle* next to the right end, in which the line reacts to its border. Everything that the reflected triangle of the line of length n learns from the half-line is written on two anti-diagonals of the half-line diagram (Lemma 3.1). Conditions on the half-line alone are cheap to check and apply to all lengths at once.

Results Our main results concern minimal-time solutions with any number of states.

(1) *The speed-*1/3 *barrier* (Theorem 3.4). Behind the front of the half-line, each row parallel to the front starts at the left border and becomes periodic after a transient. Infinitely many of these rows become periodic only after crossing the line $i = t/3$, up to an arbitrarily small angle. The proof rests on a pumping lemma (Lemma 3.3): two reflected triangles that receive the same input on the cone of their right end fire it at the same relative

Table 1: The results of this paper and the evidence behind them. A "computation" is a finite simulation or enumeration; "certified" means that every unsatisfiability claim comes with a proof that an independent checker verifies.

Result	Where	Evidence
Agreement with the half-line; return chain; determinacy	2.2, 2.4, 3.1	proof
Pumping lemma; speed-1/3 barrier	3.3, 3.4	proof
Right band	3.6	proof
Left-border lemma and barrier; no eventually regular half-line	4.1, 4.2, 4.3	proof
Finitely many early rows; the formulas $\mathcal{G}_k(T, N_1, C)$	5.3, 5.5	proof
Four states excluded by the lines of lengths $2, \dots, 5$ and parts of those of lengths $6, \dots, 10$	6.1	certified computation
Sharpness of Theorem 6.1	6.2	computation
Five rules for exactly the lengths $2, \dots, 14$	6.3	computation and proof
At least 15 neighbourhoods on the half-line of a five-state solution	6.4	certified computation and proof
The half-line of δ_{12} and δ_{13} cannot be completed	6.5	computation and proof
Balzer's conditional five-state result	6.6	certified computation

time, which is impossible for lines of different lengths. In the known constructions, the boundary of the periodic part is the signal that meets the echo at the first division point of the line; read in this way, the barrier says that this point is asymptotically never in the half of the line next to the general (Remark 3.5). In simulations, the half-line of the five-state rule δ_{12} of Section 6.3, which synchronizes the lines of lengths $2, \dots, 12$, meets the barrier exactly.

(2) *The left-border barrier* (Theorem 4.2 and Corollary 4.3). The half-line is never periodic in time and space on a wedge at its left border; in particular it is never *eventually regular*, that is, it never consists of finitely many periodic regions separated by signals of rational speeds. The reason is that the cells near the left border are fired by a finite automaton that reads the half-line along anti-diagonals, and a long periodic input would make it fire too early (Lemma 4.1). The half-line of δ_{12}, which meets the speed-1/3 barrier exactly, is excluded by this one (Corollary 6.5).

(3) *Finitely many early rows* (Corollary 5.3). The first rows of the reflected triangles of all lengths take only finitely many forms, which a certificate computed from the half-line determines. This turns the condition that no reflected triangle fires early in its first rows into finitely many checks (Lemma 5.5).

With these conditions and satisfiability solvers whose negative answers are certified by independently checked proofs, we obtain the following results. Four states are excluded by the lines of lengths $2, \dots, 5$ together with partial information on the lines of lengths $6, \dots, 10$, and this is sharp in several directions (Theorem 6.1 and Proposition 6.2). Five five-state rules synchronize exactly the lines of lengths $2, \dots, 14$ (Proposition 6.3), so the lines of lengths $2, \dots, 14$ alone cannot exclude five states. On the cells of its half-line with $t + i \leq 78$, every five-state minimal-time solution uses at least 15 distinct neighbourhoods besides the quiescent one (Proposition 6.4). Finally, a conditional five-state result reported by Balzer in 1967 holds, even in a weaker form (Proposition 6.6). Table 1 lists the results with the kind of evidence behind each of them.

Related work Indistinguishability arguments, in which a solution cannot tell two situations apart and therefore fires too early, show that there is no minimal-time solution with one-way

communication [17, Prop. 2]. Through repetitions forced by the pigeonhole principle, they also exclude minimal-time solutions for variants with sub-generals [18] and for some geometric configurations [19]. Our pumping lemma and our left-border lemma use the same mechanism to constrain solutions of the original problem. That the behaviour behind the front of a cellular automaton is eventually periodic is classical [20]; our contribution is a lower bound on where this periodicity can start. Lower bounds on the number of states are known for restricted classes and for variants such as the ring [21]; see the survey [6]. Satisfiability solvers have been applied to cellular automata before [22], and clausal proofs checked by programs independent of the solvers [23, 24] have certified large computer-assisted results in combinatorics [25, 26]. We are not aware of statements equivalent to Theorems 3.4 and 4.2 and Corollaries 4.3 and 5.3 in the literature.

Organization Section 2 introduces lines, the half-line and their agreement. Section 3 studies the right end of the reflected triangles and proves the speed-1/3 barrier, Section 4 studies their left end and proves the left-border barrier, and Section 5 reduces their first rows to finitely many cases. Section 6 applies these results to four and five states, and Section 7 discusses open questions. A list of symbols is given at the end. The programs, rule tables and logs behind every computational statement, with scripts that rerun them, are in the supplementary material, as described in the data availability statement at the end.

2 LINES AND THE HALF-LINE

2.1 Rules, Lines and Synchronization

Let S be a finite set of *states* that contains three distinct states: the *quiescent* state L, the *general* G and the *firing* state F. The other states are called *auxiliary*; the rules of this paper use A, B and, for Mazoyer's solution, C. The *working* states are $W := S \setminus \{\mathsf{F}\}$, and $* \notin S$ denotes the border. A *neighbourhood* is a triple $(x, y, z) \in (W \cup \{*\}) \times W \times (W \cup \{*\})$ with $(x, z) \neq (*, *)$, and a *rule* is a map δ from the set of neighbourhoods to S such that

$$\delta(\mathsf{L}, \mathsf{L}, \mathsf{L}) = \delta(\mathsf{L}, \mathsf{L}, *) = \mathsf{L}. \tag{1}$$

If $|S| = k$, there are $k(k-1)k - (k-1) = (k-1)^2(k+1)$ neighbourhoods: 45 for four states, 96 for five and 175 for six.

The *line of length* $n \geq 2$ is the partial map $C_n : \mathbb{N} \times \{1, \ldots, n\} \to S$ given by $C_n(0, 1) = \mathsf{G}$, $C_n(0, i) = \mathsf{L}$ for $2 \leq i \leq n$, and

$$C_n(t+1, i) = \delta\big(C_n(t, i-1),\, C_n(t, i),\, C_n(t, i+1)\big)$$

if the three arguments are defined and none of them is F; otherwise $C_n(t+1, i)$ is undefined. Here $C_n(t, 0) = C_n(t, n+1) = *$ stand for the borders. Thus a cell is defined as long as nothing in its neighbourhood has fired. The rule *synchronizes* the line of length n at time T if $C_n(T, i) = \mathsf{F}$ for $1 \leq i \leq n$ and $C_n(t, i) \in W$ for all $t < T$ and $1 \leq i \leq n$. A *minimal-time solution* is a rule that synchronizes the line of length n at time $2n - 2$ for every $n \geq 2$; it has k states if $|S| = k$.

The *half-line* $C_\infty : \mathbb{N} \times \{1, 2, \ldots\} \to S$ is defined in the same way, with $C_\infty(0, 1) = \mathsf{G}$, $C_\infty(0, i) = \mathsf{L}$ for $i \geq 2$, and a border on the left only. We draw space-time diagrams with time running downwards and the cells from left to right, as in Figure 1. The *front* is the diagonal $t = i - 1$, and the *anti-diagonal* $t + i = m$ is the set of cells (t, i) with $t + i = m$.

2.2 Agreement

Information travels at most one cell per step, so the cells that the general cannot yet have influenced stay quiescent.

Lemma 2.1 (light cone)**.** *For every rule, $C_n(t,i) = C_\infty(t,i) = \mathsf{L}$ whenever $t < i-1$ (and $i \le n$ for C_n).*

Proof. By induction on t. For $t = 0$ this is the initial condition. For $t \ge 1$, the cells $(t-1, i')$ with $i' \in \{i-1, i, i+1\}$ satisfy $t - 1 < i' - 1$, so each of them is L or the right border, and (1) gives $C_n(t,i) = C_\infty(t,i) = \mathsf{L}$. □

The right border of the line of length n is first felt by the cell n at time $n-1$, when the front reaches it, and its influence travels back to the left at speed at most one. Before this echo arrives, the line cannot be told apart from the half-line.

Lemma 2.2 (agreement)**.** *For every rule, every $n \ge 2$, $1 \le i \le n$ and $t \ge 0$ with $t + i \le 2n - 2$, the values $C_n(t,i)$ and $C_\infty(t,i)$ are both undefined or both defined and equal.*

Proof. By induction on t; the case $t = 0$ is clear. Let $t \ge 1$. If $i = n$, then $t \le n - 2 < i - 1$, and both values are L by Lemma 2.1. If $i < n$, the cells $(t-1, j)$, $j \in \{i-1, i, i+1\}$, satisfy $(t-1) + j \le 2n - 2$ and $j \le n$, so by induction they are both undefined or equal in C_n and C_∞ (for $j = 0$ both are the border), and the recursion gives the claim. □

So the border can only matter in the *reflected triangle*

$$R_n := \{(t,i) :\ 1 \le i \le n,\ t + i \ge 2n - 1,\ t \le 2n - 2\},$$

which is bounded by the anti-diagonal $t + i = 2n - 1$, the right border and the firing row $t = 2n - 2$ (Figure 1). The agreement lemma already shows why time $2n - 2$ is optimal.

Remark 2.3. Suppose that a rule synchronizes every line, the line of length n at some time $T \le 2n - 3$. Since $T + 1 \le 2n - 2$, Lemma 2.2 gives $C_\infty(T, 1) = C_n(T, 1) = \mathsf{F}$, and then $C_{n'}(T, 1) = \mathsf{F}$ for every $n' \ge n$, by Lemma 2.2 again. For $n' > T + 1$ the cell n' is still quiescent at time T by Lemma 2.1, so the line of length n' has a cell in state F at time T although not all of its cells are in state F; it can then never be synchronized, a contradiction.

2.3 The Return Chain

The first anti-diagonal of R_n is where the line starts to differ from the half-line, and in a minimal-time solution it differs there everywhere.

Lemma 2.4 (return chain)**.** *Let $n \ge 2$. If δ synchronizes the lines of lengths n and $n+1$ at times $2n-2$ and $2n$, then $C_n(2n-1-i, i) \ne C_\infty(2n-1-i, i)$ for $1 \le i \le n$, and the front cell $f_{n-1} := C_\infty(n-2, n-1)$ is not L.*

Proof. For $i = 1$ we have $C_n(2n-2, 1) = \mathsf{F}$, whereas $C_\infty(2n-2, 1) = C_{n+1}(2n-2, 1) \ne \mathsf{F}$ by Lemma 2.2 (applied to the length $n+1$) and because $2n - 2 < 2n$. Suppose that the claim holds for some $i < n$. The left and middle predecessors of the cell $(2n-1-i, i)$, namely $(2n-2-i, i-1)$ and $(2n-2-i, i)$, lie on the anti-diagonals $t + i = 2n - 3$ and $2n - 2$ (for $i = 1$ the left one is the border), so they agree in C_n and C_∞ by Lemma 2.2. As the cell itself differs, so do the right predecessors $(2n-2-i, i+1)$; all these cells are defined, since no cell of the line of length n fires before time $2n - 2$. This is the claim for $i + 1$. For $i = n$ the claim concerns the cell $(n-1, n)$. Its predecessors satisfy $C_n(n-2, n-1) = C_\infty(n-2, n-1) = f_{n-1}$ by Lemma 2.2, and the cells $(n-2, n)$ and $(n-2, n+1)$ are quiescent by Lemma 2.1, so the claim reads $\delta(f_{n-1}, \mathsf{L}, *) \ne \delta(f_{n-1}, \mathsf{L}, \mathsf{L})$. This is impossible if $f_{n-1} = \mathsf{L}$, by (1). □

We call the anti-diagonal $t + i = 2n - 1$ the *return chain* of the line of length n. Finally, a minimal-time solution never fires on the half-line: given a cell (t,i), take $n \ge t + i + 1$; then $C_\infty(t,i) = C_n(t,i)$ by Lemma 2.2, and $t < 2n - 2$, so

$$C_\infty(t,i) \in W \qquad \text{for all } t \ge 0 \text{ and } i \ge 1. \tag{2}$$

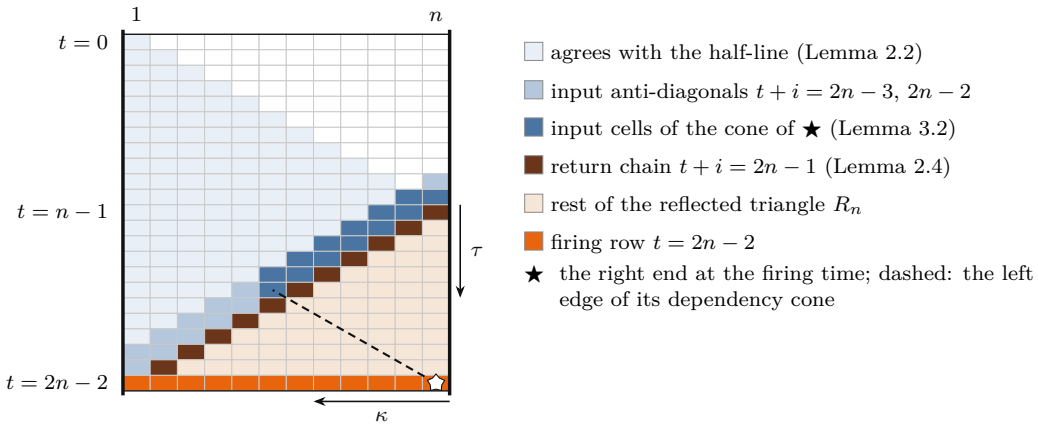


Figure 2: Anatomy of the line of length $n = 12$, coloured by region rather than by state. The upper part agrees with the half-line (Lemma 2.2). The reflected triangle R_n (Lemma 3.1) is computed from the two input anti-diagonals and the borders; its first anti-diagonal is the return chain (Lemma 2.4). The right end at the firing time (★) depends only on the dark input cells (Lemma 3.2); the dashed line is the left edge of its dependency cone. The arrows show the relative coordinates τ and κ of Section 3.

3 THE RIGHT END

The reflected triangle R_n is computed from the half-line, and its right end fires at time $2n - 2$. We show that the firing cell depends only on a cone of about $n/2$ cells of two anti-diagonals of the half-line. If two lines of different lengths received the same values on this cone, their right ends would fire at the same relative time, which is impossible (Lemma 3.3). This pumping lemma is a condition on the half-line alone, and it implies the speed-1/3 barrier (Theorem 3.4).

3.1 Relative Coordinates and Determinacy

Fix $n \geq 2$ and use the relative coordinates $\tau := t - (n-1)$ and $\kappa := n - i$: the relative time since the front reached the right end, and the distance from the right end. Write $R_n(\tau, \kappa) := C_n(n - 1 + \tau, n - \kappa)$. Then

$$R_n = \{(\tau, \kappa) : \ 0 \leq \kappa \leq \tau \leq n - 1\},$$

the right border is at $\kappa = -1$, the left border at $\kappa = n$, the return chain is the diagonal $\tau = \kappa$, and the firing row is $\tau = n - 1$ (Figure 2). The predecessors of a cell of R_n lie in R_n, on a border, or on one of the anti-diagonals $t + i = 2n - 3$ and $t + i = 2n - 2$, where the line agrees with the half-line (Lemma 2.2). The *input word* of R_n is the partial word

$$J_n(\kappa) := \big(C_\infty(n - 3 + \kappa,\, n - \kappa),\ C_\infty(n - 2 + \kappa,\, n - \kappa)\big), \qquad 0 \leq \kappa \leq n - 1,$$

whose components are the *lower* and *upper input cells*, on the anti-diagonals $t + i = 2n - 3$ and $t + i = 2n - 2$. (For $n = 2$ and $\kappa = 0$ the lower cell lies at time -1; it is a predecessor of no cell of R_n, and we set it to L.)

Lemma 3.1 (determinacy)**.** *There is a partial map Ψ, depending on δ only, such that for all $n \geq 2$ and $(\tau, \kappa) \in R_n$ the values $R_n(\tau, \kappa)$ and $\Psi(J_n, n; \tau, \kappa)$ are both undefined or both defined and equal. Moreover, $\Psi(J, m; \tau, \kappa)$ depends only on the input cells and the borders in the backward dependency cone of (τ, κ).*

Proof. Define $\Psi(J, m; \tau, \kappa)$ for $0 \leq \kappa \leq \tau \leq m - 1$ by induction on τ as $\delta(x_{\kappa+1}, x_\kappa, x_{\kappa-1})$, where each $x_{\kappa'}$ is given by the first applicable case: $*$ if $\kappa' \in \{-1, m\}$; the lower or the upper component of $J(\kappa')$ if $\tau - 1 = \kappa' - 2$ or $\tau - 1 = \kappa' - 1$, respectively; and $\Psi(J, m; \tau - 1, \kappa')$ otherwise (then $0 \leq \kappa' \leq \tau - 1$). The value Ψ is undefined if one of the arguments is F or

undefined. For $m = n$ the three arguments are the left, middle and right predecessors of the cell $(n-1+\tau, n-\kappa)$, and on the input cells C_n and C_∞ are both undefined or equal by Lemma 2.2, so induction on τ gives the first claim. Unfolding the definition gives the second. □

For $(\tau, \kappa) \in R_n$ let $\mathrm{cone}_n(\tau, \kappa)$ be the set of input cells that are reached from (τ, κ) by repeatedly passing to predecessors, stopping at input cells and borders.

Lemma 3.2 (cone of the right end)**.** *For $n \geq 2$, the cone $\mathrm{cone}_n(n-1, 0)$ avoids the left border. It consists of the upper input cell at $\kappa' = 0$, both input cells at $1 \leq \kappa' \leq \lfloor n/2 \rfloor$ and, if n is odd, the lower input cell at $\kappa' = (n+1)/2$.*

Proof. A step to a predecessor decreases τ by one and changes κ by at most one, and the input cells satisfy $\tau' \in \{\kappa' - 1, \kappa' - 2\}$. A path from $(n-1, 0)$ that reaches an input cell at κ' after r steps therefore has $\kappa' \leq r$ and $n - 1 - r \in \{\kappa' - 1, \kappa' - 2\}$, so $2\kappa' \leq n + 1$, and $2\kappa' \leq n$ for upper cells. Conversely, the upper cell at $\kappa' \leq n/2$ is reached by κ' steps to the left followed by vertical steps, and the lower cell at $1 \leq \kappa' \leq (n+1)/2$ by $\kappa' - 1$ steps to the left, vertical steps down to $(\kappa' - 1, \kappa' - 1)$ and one more step to the left, through cells of R_n; the lower cell at $\kappa' = 0$ is a predecessor of no cell of R_n. Every cell of R_n on such a path satisfies $\kappa \leq \tau \leq n - 1 - \kappa$, hence $\kappa \leq (n-1)/2$, so no such cell is adjacent to the left border $\kappa = n$. □

3.2 The Pumping Lemma

Lemma 3.3 (pumping)**.** *Let $2 \leq n < n'$, and let δ synchronize the lines of lengths n and n' at times $2n - 2$ and $2n' - 2$. Then J_n and $J_{n'}$ differ on $\mathrm{cone}_n(n-1, 0)$.*

Proof. Suppose that they agree there. The relative cell $(n-1, 0)$ lies in $R_{n'}$, and its cone in $R_{n'}$ consists of the same positions, which are away from the left border $\kappa = n'$. By Lemmas 3.1 and 3.2, the relative cell $(n-1, 0)$ of $R_{n'}$ has the value of the relative cell $(n-1, 0)$ of R_n, namely $C_n(2n-2, n) = \mathsf{F}$. Thus $C_{n'}(n' + n - 2, n') = \mathsf{F}$, although $n' + n - 2 < 2n' - 2$. □

We write $\mathrm{PUMP}(n, n')$ for the conclusion of Lemma 3.3. It is a condition on the half-line alone: the half-line must keep supplying new information to the right ends of longer and longer lines. If the depth rows that carry these cones (defined below) were periodic, the input words of two lengths that differ by a common period would agree on the cone; the next theorem turns this observation into a bound on where the periodicity can start.

3.3 Depth Rows and the Speed-1/3 Barrier

In the frame of the front, the half-line is a stack of rows parallel to the front. Let

$$\Phi_t(j) := C_\infty(t,\, t + 1 - j) \qquad (0 \leq j \leq t)$$

be the cell at *depth j* behind the front; the front cell has depth 0. The local rule becomes

$$\Phi_{t+1}(j) = \delta\big(\Phi_t(j),\, \Phi_t(j-1),\, \Phi_t(j-2)\big) \qquad (0 \leq j \leq t+1), \tag{3}$$

with $\Phi_t(-1) = \Phi_t(-2) = \mathsf{L}$ (the quiescent cells ahead of the front) and $\Phi_t(t+1) = *$ (the left border). So information only travels to larger depths, and the *depth row* $t \mapsto \Phi_t(j)$ starts at the left border at time j and then moves to the right at speed one. In these coordinates,

$$J_n(\kappa) = \big(\Phi_{n-3+\kappa}(2\kappa - 2),\ \Phi_{n-2+\kappa}(2\kappa - 1)\big) :$$

the input cells at distance κ from the right end lie on the depth rows $2\kappa - 2$ and $2\kappa - 1$.

Theorem 3.4 (speed-1/3 barrier)**.** *Let δ be a minimal-time solution. Every depth row $t \mapsto \Phi_t(j)$ is eventually periodic; let $\theta(j)$ be the least $t_0 \geq j$ such that it is periodic on $[t_0, \infty)$. For every $n \geq 4$ there is a depth $j \leq n$ with $\theta(j) > n + \frac{j}{2} - \frac{5}{2}$. Hence $\limsup_{j\to\infty} \theta(j)/j \geq 3/2$.*

Proof. Row 0 satisfies $\Phi_{t+1}(0) = \delta(\Phi_t(0), \mathsf{L}, \mathsf{L})$, so it is eventually periodic. By (3), row j is a finite-state system driven by the rows $j-1$ and $j-2$; if these are eventually periodic, so is row j. By (2) all values lie in W. Let P_j be a period of row j, fix $n \geq 4$ and let P be a common multiple of $P_0, \ldots, P_{n-1}$.

We compare the input words of R_n and R_{n+P} on $\mathrm{cone}_n(n-1, 0)$. The input cell at $\kappa = 0$ in this cone is the upper one, and it is quiescent in both triangles by Lemma 2.1. The other input cells of the cone have depth $2\kappa-2$ (lower cells) or $2\kappa-1$ (upper cells), at most $n-1$ by Lemma 3.2. Suppose that each of them lies in the periodic part of its row, that is, $n-3+\kappa \geq \theta(2\kappa-2)$ for the lower cells and $n-2+\kappa \geq \theta(2\kappa-1)$ for the upper cells. Replacing n by $n+P$ moves each input cell along its depth row by P steps, so $J_{n+P} = J_n$ on the cone, which contradicts Lemma 3.3. Hence some input cell of the cone, of depth $j \leq n$, satisfies $\theta(j) > n-3+\kappa$. For a lower cell $\kappa = (j+2)/2$ and for an upper cell $\kappa = (j+1)/2$, so in both cases $\theta(j) > n + \frac{j}{2} - \frac{5}{2}$.

Suppose now that $\limsup_{j\to\infty} \theta(j)/j < 3/2$. Then there are $\alpha < \frac{3}{2}$ and β with $\theta(j) \leq \alpha j + \beta$ for all j; since $\theta(j) \geq j$, we may take $\alpha \geq \frac{1}{2}$. For all $j \leq n$,

$$\theta(j) - \tfrac{j}{2} \leq (\alpha - \tfrac{1}{2})j + \beta \leq (\alpha - \tfrac{1}{2})n + \beta,$$

and since $\alpha - \frac{1}{2} < 1$, this is at most $n - \frac{5}{2}$ for n large. This contradicts the first part. □

In the space-time diagram, depth row j becomes periodic at the cell $(\theta(j), \theta(j)+1-j)$, and if $\theta(j) \geq (\frac{3}{2} - \varepsilon)j$, this cell lies to the right of the line $i = \frac{1-2\varepsilon}{3-2\varepsilon}\, t$ (up to an additive constant), whose slope tends to $1/3$ as $\varepsilon \to 0$. So infinitely many rows parallel to the front become periodic only after crossing the line $i = t/3$, up to an arbitrarily small angle: the transient part of the half-line, which starts at the left border, must reach this line.

Remark 3.5 (an interpretation). In the classical constructions, the line is divided recursively, and the first division point is where a signal sent by the general meets the echo of the right end. A signal of speed v that leaves the general at time 0 meets the echo, which leaves the right end at time $n-1$ at speed one, at the cell $v(2n-1)/(1+v)$. If, as for the two rules simulated below, this signal is the boundary between the transient and the periodic parts of the depth rows, then depth row j meets it at the time $(j-1)/(1-v)$, so $\theta(j) \approx j/(1-v)$. Theorem 3.4 then says that $v \geq 1/3$ asymptotically, so the first division point, at the cell $v(2n-1)/(1+v) \geq (2n-1)/4$, is asymptotically never in the half of the line next to the general. Waksman's and Balzer's solutions divide the line in the middle, with $v = 1/3$, and Mazoyer's solution divides it at two thirds of its length from the general, with $v = 1/2$. In simulations of Mazoyer's solution up to time 2600, $\theta(j) \in \{2j+1, 2j+2\}$ for $1 \leq j < 700$; for the five-state rule δ_{12} of Section 6.3, which synchronizes the lines of lengths $2, \ldots, 12$, simulations up to time 1400 give $\theta(j) = \lceil 3j/2 \rceil + 2$ for $10 \leq j \leq 300$, and $\mathrm{PUMP}(n, n')$ holds for $4 \leq n < n' \leq 650$ (Figure 3). This reading is a heuristic and not a theorem: in general, the boundary of the periodic part need not be a signal.

3.4 The Right Band

The pigeonhole argument behind the pumping lemma also works along the right border, in the direction of time. The d cells next to the right border form a finite automaton driven by the column $\kappa = d$, and this automaton has to reach its all-firing state exactly at the firing time.

Lemma 3.6 (right band)**.** *Let δ have k states and synchronize the line of length n at time $2n-2$. Let $1 \leq d \leq \tau_0 \leq n-2$ and $\ell \geq 1$, and suppose that the column $\tau \mapsto R_n(\tau, d)$ is ℓ-periodic on $[\tau_0, n-2]$, that is, $R_n(\tau+\ell, d) = R_n(\tau, d)$ whenever $\tau, \tau+\ell \in [\tau_0, n-2]$. Then $n-1-\tau_0 \leq (k-1)^d \ell$.*

Proof. For $d-1 \leq \tau \leq n-1$ put $\mathbf{b}(\tau) := (R_n(\tau, 0), \ldots, R_n(\tau, d-1))$. Since $R_n(\tau+1, \kappa) = \delta(R_n(\tau, \kappa+1), R_n(\tau, \kappa), R_n(\tau, \kappa-1))$ with $R_n(\tau, -1) = *$, there is a map Γ, depending on δ and d only, with $\mathbf{b}(\tau+1) = \Gamma(\mathbf{b}(\tau), R_n(\tau, d))$ for $d \leq \tau \leq n-2$; its arguments lie in $W^d \times W$ because

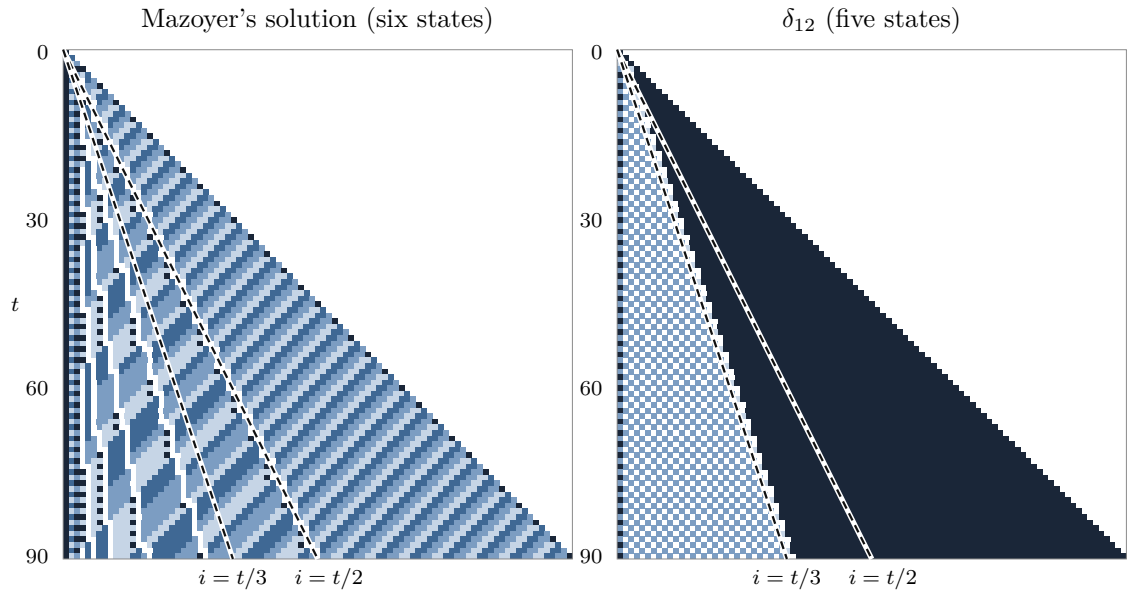


Figure 3: The half-lines of Mazoyer's six-state solution (left) and of the five-state rule δ_{12} (right) for $0 \le t \le 90$. The dashed lines are $i = t/2$ and $i = t/3$. The rows parallel to the front become periodic to the right of the line $i = t/2$ for Mazoyer's solution and of the line $i = t/3$ for δ_{12} (Remark 3.5); by Theorem 3.4, no minimal-time solution can do this to the left of the line $i = t/3$, asymptotically. To the left of the line $i = t/3$, the half-line of δ_{12} alternates between A and L, periodically in time and space. Theorem 4.2 excludes such a wedge at the left border, and Corollary 6.5 makes this explicit for the line of length 518.

no cell fires before time $2n-2$. At $\tau = n-1$ the band is F^d. Suppose that $n-1-\tau_0 > (k-1)^d\ell$. The pairs $(\mathbf{b}(\tau), \tau \bmod \ell)$ with $\tau_0 \le \tau \le n-2$ take at most $(k-1)^d\ell$ values, so there are $\tau_0 \le \tau_1 < \tau_2 \le n-2$ with $\mathbf{b}(\tau_1) = \mathbf{b}(\tau_2)$ and $\ell \mid h := \tau_2 - \tau_1$. For $0 \le s \le n-2-\tau_2$ the driving cells $R_n(\tau_1 + s, d)$ and $R_n(\tau_2 + s, d)$ lie in $[\tau_0, n-2]$ and their times differ by a multiple of the period, so they are equal, and induction on s gives $\mathbf{b}(\tau_1 + s) = \mathbf{b}(\tau_2 + s)$ for $0 \le s \le n-1-\tau_2$. For $s = n-1-\tau_2$ this reads $\mathbf{b}(n-1-h) = \mathbf{b}(n-1) = \mathsf{F}^d$, although $n-1-h < n-1$, a contradiction. □

4 THE LEFT END

Near the left border, the reflected triangle is produced by a band of anti-diagonals that follow the echo of the right end, the *chains*. The first few chains form a finite automaton that reads the two input anti-diagonals and must fire the cells near the left border exactly when the echo reaches them. If the input anti-diagonals were periodic over a long stretch, this automaton would repeat a state and fire too early (Lemma 4.1). Applied to all lengths at once, this excludes periodic wedges at the left border of the half-line (Theorem 4.2).

4.1 Chains and Slices

Fix $n \ge 2$. For $c \ge 0$ and $\max(1, c) \le i \le n$, and for $c = -1$ and $1 \le i \le n-1$, put

$$v_c(i) := C_n(2n-2+c-i,\, i),$$

the cell at position i of the *chain* $t + i = 2n-2+c$. Chains -1 and 0 are the input anti-diagonals, and chain 1 is the return chain (Figure 4). The predecessors of the cell $(2n-2+c-i, i)$ lie on the chains $c-2$, $c-1$ and c, so

$$v_c(i) = \delta(v_{c-2}(i-1),\, v_{c-1}(i),\, v_c(i+1)) \qquad (c \ge 1), \tag{4}$$

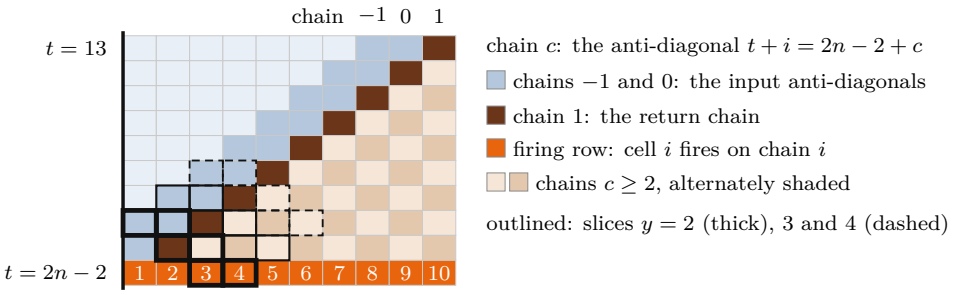


Figure 4: The lower left corner of the line of length $n = 12$, coloured by chain. The cell i of the firing row lies on chain i. Each outlined group of six cells is a slice $(u_1(y), \dots, u_4(y))$ of the band of chains $1, \dots, 4$ used in the proof of Lemma 4.1 with $s = 2$, together with its two input cells $u_{-1}(y)$ and $u_0(y)$: the slice is computed from the slice to its right and from these two input cells. The group drawn with a thick line is the one at $y = s$; its last cell $u_{2s}(s)$ is in the firing row.

where $v_{c-2}(0)$ and $v_c(n+1)$ stand for the borders. If the line of length n is synchronized at time $2n-2$, then the cell i fires on chain i: $v_i(i) = \mathsf{F}$, and $v_c(i) \neq \mathsf{F}$ for $c < i$. In the skewed coordinates $u_c(y) := v_c(y + \lfloor c/2 \rfloor)$ the recursion (4) becomes

$$u_c(y) = \delta(u_{c-2}(y),\, u_{c-1}(y + \varepsilon_c),\, u_c(y+1)), \qquad \varepsilon_c = 1 \text{ for even } c, \quad \varepsilon_c = 0 \text{ for odd } c, \quad (5)$$

with $u_0(y) = C_\infty(2n-2-y, y)$ and $u_{-1}(y) = C_\infty(2n-2-y, y-1)$. Here $u_c(y)$ is the cell $y + \lfloor c/2 \rfloor$ at time $2n - 2 + \lceil c/2 \rceil - y$; it is F if $\lceil c/2 \rceil = y$ and not F if $\lceil c/2 \rceil < y$. So each *slice* $(u_1(y), u_2(y), \dots)$ is computed from the slice $y+1$ and the two input cells $u_{-1}(y)$ and $u_0(y)$: reading the input anti-diagonals from right to left, the slices form a finite automaton. A word w on an integer interval I is *ℓ-periodic* if $w(y+\ell) = w(y)$ whenever $y, y+\ell \in I$.

4.2 The Left-Border Lemma

Lemma 4.1 (left border)**.** *Let δ have k states and synchronize the line of length n at time $2n-2$. Let $s, \ell \geq 1$ and $s \leq Y \leq n - s$, and suppose that the words $y \mapsto C_\infty(2n-2-y, y)$ for $s \leq y \leq Y$ and $y \mapsto C_\infty(2n-3-y, y)$ for $s \leq y \leq Y-1$ are ℓ-periodic. Then $Y - s \leq (k-1)^{2s}\ell$.*

Proof. For $s \leq y \leq Y$ let $\mathbf{u}(y) := (u_1(y), \dots, u_{2s}(y))$. These are the cells $y + \lfloor c/2 \rfloor \leq n$ at the times $2n - 2 + \lceil c/2 \rceil - y$ with $\lceil c/2 \rceil \leq s \leq y$, so they are defined, and none of them is F when $y > s$. By (5), $u_c(y)$ with $1 \leq c \leq 2s$ is computed from components of index less than c of $\mathbf{u}(y)$, components of index at most c of $\mathbf{u}(y+1)$, and the input cells $u_{-1}(y)$ and $u_0(y)$. Hence $\mathbf{u}(y) = \Gamma(\mathbf{u}(y+1), u_{-1}(y), u_0(y))$ for $s+1 \leq y \leq Y-1$, with Γ depending on δ only. Unfolding (5) for even c, $u_{2j}(y) = \delta(u_{2j-2}(y), u_{2j-1}(y+1), u_{2j}(y+1))$, shows that $u_{2s}(y) = \Gamma'(\mathbf{u}(y+1), u_0(y))$ for $s \leq y \leq Y-1$. By hypothesis, u_0 is ℓ-periodic on $[s, Y]$ and u_{-1} on $[s+1, Y]$.

Suppose that $Y - s > (k-1)^{2s}\ell$. The pairs $(\mathbf{u}(y), y \bmod \ell)$ with $s+1 \leq y \leq Y$ take at most $(k-1)^{2s}\ell$ values, so there are $s+1 \leq y_1 < y_2 \leq Y$ with $\mathbf{u}(y_1) = \mathbf{u}(y_2)$ and $\ell \mid h := y_2 - y_1$. By downward induction, $\mathbf{u}(y) = \mathbf{u}(y+h)$ for $s+1 \leq y \leq y_1$: if this holds for $y+1 \leq y_1$, then

$$\mathbf{u}(y) = \Gamma(\mathbf{u}(y+1), u_{-1}(y), u_0(y)) = \Gamma(\mathbf{u}(y+1+h), u_{-1}(y+h), u_0(y+h)) = \mathbf{u}(y+h),$$

since $y + h < y_2 \leq Y$. As $s + h \leq Y - 1$ and $u_0(s+h) = u_0(s)$,

$$u_{2s}(s+h) = \Gamma'(\mathbf{u}(s+1+h), u_0(s+h)) = \Gamma'(\mathbf{u}(s+1), u_0(s)) = u_{2s}(s) = \mathsf{F},$$

although $\lceil 2s/2 \rceil = s < s + h$, so that $u_{2s}(s+h)$, the cell $2s+h$ at time $2n-2-h$, is not F. This is a contradiction. □

In words: if the two input anti-diagonals are ℓ-periodic over a stretch of more than $(k-1)^{2s}\ell$ cells that starts at the cell s, then the band of the first $2s$ chains repeats a state, and the cell $2s+h$ fires h steps too early. The bound is exponential in s, but for $s=1$ and $s=2$ it is small enough to be useful in computations (Proposition 6.4 and Corollary 6.5).

4.3 No Periodic Wedge at the Left Border

Theorem 4.2 (left-border barrier)**.** *Let δ be a minimal-time solution. There are no $s \geq 1$, $\alpha > 0$, t_0 and $p, q \geq 1$ such that $C_\infty(t+p,i) = C_\infty(t,i)$ and $C_\infty(t,i+q) = C_\infty(t,i)$ whenever the cells involved lie in the wedge*

$$\Omega := \{(t,i) : \ s \leq i \leq \alpha t, \ t \geq t_0\}.$$

Proof. Suppose otherwise and put $\ell := pq$. If (t,i) and $(t-\ell, i+\ell)$ lie in Ω, so do the cells $(t-\ell, i+jq)$ for $0 \leq j \leq p$ and $(t-\ell+jp, i)$ for $0 \leq j \leq q$, hence $C_\infty(t-\ell, i+\ell) = C_\infty(t-\ell, i) = C_\infty(t,i)$. Let $Y_n := \lfloor \alpha(2n-3)/(1+\alpha) \rfloor$. For $s \leq y \leq Y_n$ the cells $(2n-2-y, y)$ and $(2n-3-y, y)$ satisfy $y \leq \alpha(2n-3-y)$, so they lie in Ω as soon as $2n-3-Y_n \geq t_0$, that is, for all large n. A step of ℓ positions along an anti-diagonal is the step from (t,i) to $(t-\ell, i+\ell)$, so the words of Lemma 4.1 are ℓ-periodic on $[s, Y_n]$ and $[s, Y_n - 1]$. As $Y_n \geq \frac{2\alpha}{1+\alpha} n - 4$, for large n the number $Y := \min(Y_n, n-s)$ satisfies $Y - s > (k-1)^{2s}\ell$, which contradicts Lemma 4.1. □

The half-line of a rule is *eventually regular* if there are t_1, $P \geq 1$ and, for each residue $r \in \{0, \ldots, P-1\}$, an integer m, words $w_0, \ldots, w_m$, nonempty words $z_1, \ldots, z_m$ over W and integers $a_j, D_j \geq 0$ (all depending on r) such that for all $\nu \geq 0$ the configuration $C_\infty(t,1) \cdots C_\infty(t,t+1)$ at the time $t = t_1 + r + \nu P$ is $w_0 z_1^{a_1+\nu D_1} w_1 \cdots z_m^{a_m+\nu D_m} w_m$. In words: from some time on, the half-line consists of finitely many periodic regions separated by signals of rational speeds; on signals in one-dimensional cellular automata see [20]. The half-line of δ_{12} in Figure 3 looks like this: a front, a region of G's behind it and, beyond a boundary of speed 1/3, a region that alternates between A and L.

Corollary 4.3. *The half-line of a minimal-time solution is not eventually regular.*

Proof. Suppose that it is. For each residue r some D_j is positive, because the configuration grows by P cells per period; let $j(r)$ be the least such index. The blocks and the junctions to the left of $z_{j(r)}$ have constant lengths, so the block $z_{j(r)}^{a_{j(r)}+\nu D_{j(r)}}$ starts at a fixed cell s_r, and its length is at least βt for some $\beta > 0$ and all large t of the residue class r. Inside this block, the state of a cell depends only on its distance to s_r modulo $|z_{j(r)}|$, and it is the same at the times t and $t+P$. Let $s := \max_r s_r$, $q := \prod_r |z_{j(r)}|$ and $\alpha := \beta/2$. For t_0 large, every cell of the wedge $\{(t,i) : s \leq i \leq \alpha t, \ t \geq t_0\}$ lies in the block of index $j(r)$ of its residue class, so C_∞ is periodic with period P in time and q in space on this wedge. This contradicts Theorem 4.2. □

5 THE FIRST ROWS AFTER THE REFLECTION

The five-state rules that come closest to minimal time fail in the first rows of their reflected triangles: some cell of a long line fires a few steps after the front has reached the right end (Proposition 6.3 and Figure 5). The condition that this never happens concerns infinitely many lengths. We show that, for a given half-line, it reduces to finitely many lengths, and that a certificate for the reduction can be computed by simulating the half-line.

5.1 Finitely Many Early Rows

For $T \geq 0$ let $E(T) := \{(\tau, \kappa) : 0 \leq \kappa \leq \tau \leq T\}$ be the first $T+1$ rows of a reflected triangle. If $n \geq T+2$, then $E(T) \subseteq R_n$, and no cell of $E(T)$ lies in the firing row $\tau = n-1$.

Lemma 5.1 (early rows)**.** *Let δ be a rule, $T \geq 0$, and $n, n' \geq T+2$ such that the partial values $J_n(\kappa)$ and $J_{n'}(\kappa)$ coincide for $0 \leq \kappa \leq T+1$. Then for every $(\tau, \kappa) \in E(T)$ the values $R_n(\tau, \kappa)$ and $R_{n'}(\tau, \kappa)$ are both undefined or both defined and equal.*

Proof. By Lemma 3.1 we may compare $\Psi(J_n, n; \tau, \kappa)$ and $\Psi(J_{n'}, n'; \tau, \kappa)$. Unfolding the recursion of Lemma 3.1 from $(\tau, \kappa) \in E(T)$ visits only pairs (τ'', κ'') with $0 \leq \kappa'' \leq \tau'' \leq T$ and reads the symbols $x_{\kappa'}$ with $\kappa' \leq \kappa'' + 1 \leq T+1$. The case $\kappa' = m$ would require $m \leq T+1$, which is excluded for $m \in \{n, n'\}$; hence the left border is never read, and the input word is read only at positions $\kappa' \leq T+1$, where J_n and $J_{n'}$ coincide. The two recursions are therefore identical step by step, including the points where an argument is F or undefined. □

The input word on $[0, T+1]$ consists of depth-row cells of depths at most $2T+1$. These rows are driven only by the rows of smaller depth (3), so the vector of the first $2T+2$ depth rows evolves autonomously, and one period of it is a certificate for all later times.

Lemma 5.2 (periodicity certificate)**.** *Let δ be a rule, $D \geq 0$, and $V_t := (\Phi_t(0), \ldots, \Phi_t(D))$ for $t \geq D$. There is a map G_D on W^{D+1}, depending on δ and D only, with $V_{t+1} = G_D(V_t)$ whenever $t \geq D$ and $V_t \in W^{D+1}$. If $V_{t_0} = V_{t_0+p}$ for some $t_0 \geq D$ and $p \geq 1$, and $V_t \in W^{D+1}$ for $t_0 \leq t < t_0 + p$, then $V_t \in W^{D+1}$ and $V_{t+p} = V_t$ for all $t \geq t_0$.*

Proof. For $t \geq D$ and $0 \leq j \leq D$ we have $j \leq t$, so the arguments in (3) are $\Phi_t(j)$, $\Phi_t(j-1)$ and $\Phi_t(j-2)$, with indices at most D, and L for negative indices; none of them is the border symbol $\Phi_t(t+1) = *$. Hence $G_D(v)_j := \delta(v_j, v_{j-1}, v_{j-2})$, with $v_{-1} = v_{-2} = \mathsf{L}$, has the required property. For the second statement, suppose that $V_{t'} \in W^{D+1}$ and $V_{t'+p} = V_{t'}$ for $t_0 \leq t' \leq t$, which is true for $t = t_0$. Then $V_{t+1+p} = G_D(V_{t+p}) = G_D(V_t) = V_{t+1}$, and $V_{t+1} \in W^{D+1}$, by hypothesis if $t+1 < t_0 + p$ and because $V_{t+1} = V_{t+1-p}$ otherwise. Induction on t concludes. □

Corollary 5.3 (finitely many early rows)**.** *Let δ, $T \geq 0$, $D := 2T+1$, $t_0 \geq D$ and $p \geq 1$ satisfy the hypothesis of Lemma 5.2. Then $J_n(\kappa) = J_{n+p}(\kappa) \in W^2$ for all $0 \leq \kappa \leq T+1$ and $n \geq t_0 + 3$. Hence, for every $N_0 \geq T+2$, every R_n with $n \geq N_0$ coincides on $E(T)$, in the sense of Lemma 5.1, with $R_{n'}$ for one of the finitely many lengths $N_0 \leq n' < \max(N_0, t_0+3) + p$.*

Proof. For $1 \leq \kappa \leq T+1$ the components of $J_n(\kappa)$ are cells of the depth rows $2\kappa - 2$ and $2\kappa - 1 \leq D$ at the times $n-3+\kappa$ and $n-2+\kappa$, which are at least $n-2 > t_0$; by Lemma 5.2 they lie in W and do not change when n is replaced by $n+p$. For $\kappa = 0$ the lower component is L, by convention or by Lemma 2.1, and the upper one is $C_\infty(n-2, n) = \mathsf{L}$. If $n \geq \max(N_0, t_0+3) + p$, subtracting a multiple of p gives a length n' in $[\max(N_0, t_0+3), \max(N_0, t_0+3) + p)$ with $J_{n'} = J_n$ on $[0, T+1]$, and Lemma 5.1 applies since $n, n' \geq T+2$. □

For a minimal-time solution, the depth rows are eventually periodic (Theorem 3.4), so t_0 and p as in Corollary 5.3 always exist. Thus the condition that no reflected triangle R_n with $n \geq N_0$ has a firing cell in its first $T+1$ rows reduces to finitely many lengths, with a certificate (t_0, p) found by simulating the half-line.

5.2 Early-Row Formulas

For a search over all rules, the half-line is not known in advance, but its first rows and the lines of small lengths can be encoded together. The following conditions collect what a minimal-time solution must satisfy on the complete lines of small lengths, on the early rows and return chains of the next lengths, and on the cone of their right ends.

Definition 5.4. Let $k \geq 3$, $T \geq 1$, $N_1 \geq T+2$ and $T+1 \leq C \leq N_1$. A k-state rule δ satisfying (1) *satisfies* $\mathcal{G}_k(T, N_1, C)$ if

(G1) δ synchronizes the line of length n at time $2n-2$ for $2 \leq n \leq T+1$;

(G2) for $T+2 \le n \le N_1$, $1 \le i \le n$ and $t \le n-1+T$: $C_n(t,i) \in W$;

(G3) for $T+2 \le n \le N_1$ and $1 \le i \le n$ with $t := 2n-1-i \le n-1+T$: $C_n(t,i) \neq C_\infty(t,i)$;

(G4) $C_\infty(t,i) \in W$ whenever $t+i \le 2N_1$;

(G5) for $T+2 \le n \le C$: $R_n(\tau,\kappa) \in W$ for all $(\tau,\kappa) \in R_n$ with $\tau+\kappa \le n-1$ and $\tau < n-1$, $R_n(n-1,0) = \mathsf{F}$, and $R_n(\kappa,\kappa) \neq C_\infty(n-1+\kappa, n-\kappa)$ for $2\kappa \le n-1$.

We write $\mathcal{G}_k(T,N_1)$ for $\mathcal{G}_k(T,N_1,T+1)$, in which (G5) is void.

Condition (G2) says that no cell of the lines of lengths $T+2,\dots,N_1$ fires up to the relative time $\tau = T$, and (G3) is the return chain of Lemma 2.4, restricted to the relative times $\tau \le T$. Condition (G5) concerns the backward dependency cone $\{(\tau,\kappa) \in R_n : \tau+\kappa \le n-1\}$ of the firing cell $(n-1,0)$; by the local definition of C_n, its cells are defined even if a cell outside the cone fires earlier.

Lemma 5.5. *Every k-state minimal-time solution satisfies $\mathcal{G}_k(T,N_1,C)$ for all T, N_1 and C as in Definition 5.4.*

Proof. (G1) holds by definition. For (G2), $n \ge T+2$ gives $n-1+T < 2n-2$, and every cell of the line of length n is in W before time $2n-2$. (G3) is Lemma 2.4, which applies because the lines of lengths n and $n+1$ are synchronized at times $2n-2$ and $2n$. (G4) is (2). For (G5), the cells of R_n with $\tau < n-1$ are in W, $R_n(n-1,0) = C_n(2n-2,n) = \mathsf{F}$, and the return chain differs from C_∞ by Lemma 2.4. □

For a fixed half-line, the early rows of all lengths $n \ge N_0$ can thus be checked on the representatives of Corollary 5.3; Proposition 6.3 is proved in this way, and the same representatives allow a formula to impose (G2) and (G3) on all $n \ge N_0$.

6 APPLICATIONS TO FOUR AND FIVE STATES

We now combine the necessary conditions with satisfiability solvers. All statements of this section are computer-assisted. Each of them is either a finite simulation or enumeration, or the unsatisfiability of a propositional formula that is certified by a proof checked independently of the solver; the kind of evidence is stated for each result and summarized in Table 1.

6.1 Method

Encodings The formula $\mathrm{MT}_k(\mathcal{N}; M, N')$ has one-hot variables for the entries of a k-state rule table, for the cells of C_∞ with $t+i \le M$ and for the cells of R_n with $n \in \mathcal{N}$. Its clauses state the local rule, that no half-line cell is F, that the line of length n is synchronized at time $2n-2$ for every $n \in \mathcal{N}$, the return chains of Lemma 2.4 for $n, n+1 \in \mathcal{N}$, and $\mathrm{PUMP}(n,n')$ for $4 \le n < n' \le N'$; the formula $\mathrm{MT}_k(\mathcal{N}; M)$ has no pumping clauses. The formula $\mathcal{G}_k(T,N_1,C)$ is encoded in the same way, with $M = 2N_1$. Its regions are closed under predecessors, so the encoding is equivalent to Definition 5.4.

Certificates Every unsatisfiability claim is certified by a proof in the LRAT format [24], which is produced by the solver CaDiCaL [27] or, for one formula, obtained from a DRAT proof of the solver Kissat [28] with `drat-trim` [23], and verified by the independent checker `lrat-check`. The formula generators, the enumerations and the simulations are trusted programs, and every model found by a solver was re-simulated by a checker that shares no code with the generators.

Validation As tests of the encodings, Mazoyer's solution, as extracted from Duprat's formal proof of its correctness [29], satisfies $\mathrm{MT}_6(\{3,\dots,12\};78,40)$ and, with its table fixed, (G2) and (G3) for $T = 8$ and all $n \geq 15$. With the table of the rule δ_{14} of Section 6.3 fixed, the formulas are satisfiable for the lines of lengths $2,\dots,14$ and unsatisfiable with the line of length 15, or with the early rows ($T = 8$, but not $T = 6$) of the lines of lengths $15,\dots,30$, as simulation predicts.

Reproducibility The supplementary material contains the programs, the rule tables, the lists of half-line germs and the logs, with two scripts that rerun every computation: one for the simulations, enumerations and satisfiable formulas (about 7 minutes), and one that regenerates and checks every LRAT proof (about 30 minutes). The solver times, proof sizes and SHA-256 digests of all formulas and proofs are listed there; see the data availability statement at the end.

6.2 Four States

Theorem 6.1. *No four-state rule satisfying* (1)

(i) synchronizes all lines of lengths $2,\dots,9$ *in minimal time;*

(ii) satisfies $\mathcal{G}_4(6,10)$*: the lines of lengths* $2,\dots,7$ *complete and, for the lines of lengths* 8, 9 *and* 10*, the first seven rows and the return chains;*

(iii) satisfies $\mathcal{G}_4(4,10,10)$*: the lines of lengths* $2,\dots,5$ *complete and, for the lines of lengths* $6,\dots,10$*, the first five rows, the return chains and the right ends.*

In particular, no four-state minimal-time solution exists.

Proof. (i) Such a rule yields a model of $\mathrm{MT}_4(\{2,\dots,9\};16)$: a half-line cell with $t + i \leq 16$ is quiescent if $i \geq 10$ (Lemma 2.1) and equals a cell of the line of length 9 at a time $t < 16$ if $i \leq 9$ (Lemma 2.2), so it is in W; and the return chains hold by Lemma 2.4 for $2 \leq n \leq 8$. This formula has 753 variables and 17,605 clauses. (ii) The encoding of $\mathcal{G}_4(6,10)$ has 919 variables and 22,960 clauses. (iii) The encoding of $\mathcal{G}_4(4,10,10)$ has 874 variables and 20,461 clauses. CaDiCaL refutes each of the three formulas with an LRAT proof, and `lrat-check` verifies each proof. The last claim follows from (i), and also from (ii) or (iii) with Lemma 5.5. □

Part (i) certifies the bound observed by Yunès [10]; the earlier proofs [7, 10, 11] rest on the correctness of search programs. Parts (ii) and (iii) show how little information excludes four states: in (iii), only the lines of lengths up to 5 are complete. These bounds are sharp in the following sense.

Proposition 6.2 (computational)**.**

(a) Exactly 27 *partial four-state rules (rules restricted to the neighbourhoods they use) synchronize all lines of lengths* $2,\dots,8$ *in minimal time, and no partial four-state rule synchronizes all lines of lengths* $2,\dots,9$*.*

(b) The formulas $\mathcal{G}_4(5,30)$*,* $\mathcal{G}_4(6,9)$*,* $\mathcal{G}_4(4,30,9)$ *and* $\mathcal{G}_4(3,30,12)$*, the formula* $\mathcal{G}_4(6,10)$ *without (G3), and the formula that consists of (G1) for* $T = 4$ *and (G5) for* $6 \leq n \leq 12$ *are satisfiable.*

Proof. (a) is an exhaustive enumeration. Yunès [10] found the same number under the extra condition $\delta(*,\mathsf{L},\mathsf{L}) = \mathsf{L}$, which is vacuous here because none of the 27 partial rules uses the neighbourhood $(*,\mathsf{L},\mathsf{L})$. (b) The solver finds a model of each formula, and each model is re-simulated. □

So in part (ii) of Theorem 6.1 neither T nor N_1 can be lowered and (G3) cannot be dropped; in part (iii) neither T nor C can be lowered, even with more early rows, and neither the early rows nor the right ends suffice on their own.

Table 2: The five-state rule δ_{14}. The entry in row (x, y) and column z is $\delta_{14}(x, y, z)$; the neighbourhoods $(*, y, *)$ are not used ("–").

x	y	$z=$ L	G	A	B	$*$	x	y	$z=$ L	G	A	B	$*$
$*$	L	F	F	L	G	–	$*$	A	G	G	L	G	–
L	L	L	G	G	B	L	L	A	B	G	B	A	A
G	L	A	B	A	B	G	G	A	A	G	L	A	B
A	L	B	G	G	G	G	A	A	L	A	G	A	B
B	L	B	A	A	L	A	B	A	A	A	G	A	L
$*$	G	L	F	L	L	–	$*$	B	G	G	G	G	–
L	G	A	F	B	A	F	L	B	B	G	A	L	G
G	G	B	F	G	F	B	G	B	B	A	B	B	F
A	G	A	G	G	A	B	A	B	B	B	G	B	G
B	G	A	G	A	L	B	B	B	L	A	A	L	A

6.3 Five States: Rules for the Lengths Up to Fourteen

The following five-state rules were found with satisfiability solvers. Their tables are the files `uniformA_2-12.txt` (the rule δ_{12}), `delta13.txt` and `delta14*.txt` in `results/` of the supplementary material. Table 2 shows δ_{14}.

Proposition 6.3. *Each of the five-state rules δ_{14}, δ_{14}^{b}, δ_{14}^{c}, δ_{14}^{d} and δ_{14}^{e} synchronizes the line of length n at time $2n-2$ for $2 \le n \le 14$, and for no $n \ge 15$. More precisely, for every $n \ge 15$, some cell of the line of length n is* F *at a relative time $\tau \le \min(T, n-2)$, where $T = 10, 14, 22, 12, 14$, respectively.*

Proof. The lines of lengths $2, \ldots, 60$ are simulated. For $15 \le n \le 60$ some cell is F at a relative time $\tau \le n-2$, which proves the claim for $15 \le n < T+3$. The half-line has no cell in state F at the times $t < 3000$, so by Lemma 2.2 a cell of a line of length $n \le 1500$ that is F at a time $t \le 2n-2$ lies in R_n. Simulating the half-line gives t_0 and p that satisfy the hypothesis of Lemma 5.2 for $D = 2T+4 \ge 2T+1$, namely $(t_0, p) = (51, 8)$, $(66, 3)$, $(257, 192)$, $(95, 48)$ and $(56, 2)$. For every n with $\max(15, T+3) \le n < t_0 + 3 + p \le 452$, direct simulation shows that the line of length n, or of a length n' in this range with $J_{n'} = J_n$ on $[0, T+1]$, has a cell in state F at a relative time $\tau \le T$, hence in $E(T)$. (One length is simulated for each class of a partition that is finer than equality of J_n on $[0, T+1]$; together with the lengths below $T+3$, these are 35, 39, 413, 116 and 28 lengths, the largest being 49, 53, 427, 130 and 42.) By Lemma 5.1 and Corollary 5.3 with $N_0 = \max(15, T+3)$, every R_n with $n \ge N_0$ has a cell of $E(T)$ in state F, and there $\tau \le T \le n-3$. □

Figure 5 shows how δ_{14} fails. For $3 \le t < 1500$ its half-line takes the values L, G at cell 1, L, A at cell 2 and L, B at the cells $i \ge 3$, and δ_{14} restricted to $\{\mathsf{L}, \mathsf{B}\}^3$ is the elementary rule 22 [30]. In the simulated ranges, neither barrier excludes this half-line: PUMP(n, n') holds for $4 \le n < n' \le 800$, and Lemma 4.1 does not apply for $n \le 5000$, $s \le 3$ and $\ell \le 24$. The rules δ_{12} and δ_{13} synchronize the lines of lengths $2, \ldots, 12$ and $2, \ldots, 13$, respectively, and no other line of length at most 60. Each of these seven rules has a line of some length $14 \le n \le 19$ with a cell in state F at a relative time $\tau \le 9$, so none of them satisfies $\mathcal{G}_5(9, N_1)$ for $N_1 \ge 19$.

6.4 Five States: Half-Lines of Low Complexity

The barriers constrain the half-line, so it is natural to ask how simple the half-line of a five-state minimal-time solution can be. We measure this by the number of distinct neighbourhoods that it uses in a fixed region. The *germ* below M of a rule is its restriction to the neighbourhoods of the cells (t, i) of its half-line with $t \ge 1$ and $t + i \le M$, and its *complexity* $c_M(\delta)$ is the

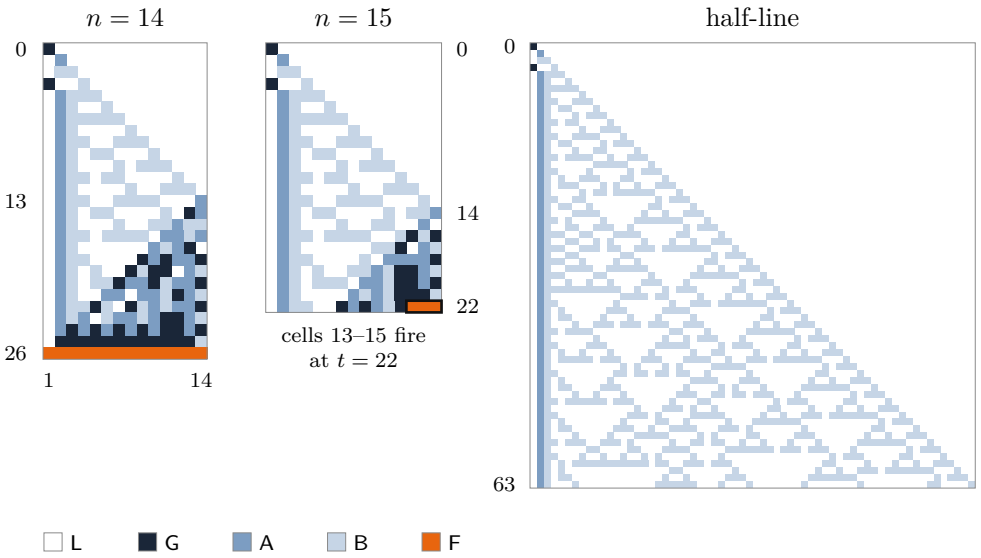


Figure 5: The rule δ_{14}. Left: the line of length 14 is synchronized at time 26. Middle: on the line of length 15, the cells 13, 14 and 15 fire at time 22, that is, at the relative time $\tau = 8$, six steps early. Right: the half-line for $0 \le t \le 63$. From time 3 on, the cells $i \ge 3$ take only the values L and B, and there δ_{14} acts as the elementary cellular automaton 22.

number of these neighbourhoods other than (L, L, L). We use $M = 78$: the half-line below the anti-diagonal $t + i = 78$ contains the input anti-diagonals of all lines of lengths $n \le 40$, so the germ below 78 determines the conditions PUMP(n, n') for $n' \le 40$. For example, $c_{78} = 57$ for Mazoyer's solution, 22 for δ_{12} and 16 for δ_{14}.

Proposition 6.4. *Every five-state minimal-time solution δ satisfies $c_{78}(\delta) \ge 15$.*

Proof. Suppose that $c_{78}(\delta) \le 14$. Exchanging A and B if necessary, we may assume that A occurs before B (if at all) in the order of anti-diagonals and, within an anti-diagonal, of time. By (2) and Lemmas 2.4 and 3.3, the germ of δ below 78 produces no F, never produces L at the front, and satisfies PUMP(n, n') for $4 \le n < n' \le 40$. A depth-first search that chooses the value of each neighbourhood at its first occurrence enumerates all such germs, visiting about 10^8 nodes: there are 244, and an independent re-implementation finds the same 244. For 230 of them, the formula $\mathrm{MT}_5(\{2, \dots, 10\}; 78, 39)$ with the germ fixed is unsatisfiable. A single formula covers all 230: it has a selector variable for each germ, clauses that make each selector fix its germ, and a clause that requires some selector; CaDiCaL refutes it with an LRAT proof that `lrat-check` verifies. Each of the other 14 germs determines C_∞ up to the anti-diagonal 3300, and there Lemma 4.1 applies with $s = 1$: for some $n \le 163$ and $\ell \le 5$, the upper input anti-diagonal of the line of length n is ℓ-periodic on $[1, Y]$ and the lower one on $[1, Y - 1]$ with $Y - 1 > 16\ell$ (for example $n = 28$, $\ell = 1$ and $Y = 19$), as an independent simulation confirms. Hence no rule that contains one of the 244 germs is a minimal-time solution. □

The bound is attained by rules that come close to minimal time: the germs below 78 of δ_{14}^{c} and δ_{14}^{d} have complexity 15, and both rules synchronize the lines of lengths $2, \dots, 14$ (Proposition 6.3).

The rules δ_{12} and δ_{13} have the same half-line on the cells with $t + i \le 78$, where they use the 22 neighbourhoods

```
*GL>A GLL>G *AG>B AGL>L *BL>A BLG>L LGL>B *AL>G ALB>A LBG>L BGL>G
*GA>A GAL>L ALG>B LGG>G GGL>G BGG>L GGG>G ALL>A LLG>L ALA>A LAL>L
```

besides (L, L, L); here `xyz>d` means $\delta(x, y, z) = d$. This half-line consists of a front, a region of G's behind it and, beyond a boundary of speed 1/3, a region that alternates between A and L,

periodically in time and space (Figure 3). By Theorem 4.2 no minimal-time solution has such a half-line, and Lemma 4.1 gives an explicit length.

Corollary 6.5. *No five-state rule that has the* 22 *transitions above synchronizes the line of length* 518 *in minimal time.*

Proof. With these transitions and $\delta(\mathsf{L},\mathsf{L},\mathsf{L}) = \mathsf{L}$, no other neighbourhood occurs on the cells with $t + i \le 1034$, so every such rule has the same diagram C_∞ there. On the anti-diagonal $t+i = 1034$ the cells $2, \dots, 259$ are in state A, and on $t+i = 1033$ the cells $2, \dots, 258$ are in state L. Lemma 4.1 with $k = 5$, $n = 518$, $s = 2$, $\ell = 1$ and $Y = 259$ would give $257 \le 4^4 = 256$. □

6.5 Balzer's Conditions

Balzer [7, p. 37] reported that no five-state minimal-time solution satisfies the following four conditions:

(B1) $\delta(x, \mathsf{G}, y) = \mathsf{G}$ unless $x, y \in \{\mathsf{G}, *\}$;

(B2) $\delta(I(z), I(y), I(x)) = I(\delta(x, y, z))$ for an involution I of the states with $I(*) = *$;

(B3) $\delta(\mathsf{G},\mathsf{G},\mathsf{G}) = \delta(*,\mathsf{G},\mathsf{G}) = \delta(\mathsf{G},\mathsf{G},*) = \mathsf{F}$, and no other neighbourhood gives F;

(B4) $\delta(\mathsf{G}, v, \mathsf{G}) = \mathsf{G}$ for all $v \in W \setminus \{\mathsf{G}\}$.

In the *weak reading* [10], (B1) is required only for working states x, y, and (B3) without its last clause. We take $I(\mathsf{F}) = \mathsf{F}$. Then $I(\mathsf{G}) = \mathsf{G}$: otherwise $g := I(\mathsf{G}) \in W \setminus \{\mathsf{G}\}$, and (B2) and (B4) give $\delta(g, \mathsf{G}, g) = I(\delta(\mathsf{G}, g, \mathsf{G})) = I(\mathsf{G}) = g$, whereas (B1) gives $\delta(g, \mathsf{G}, g) = \mathsf{G}$. So I is one of the four involutions of $\{\mathsf{L}, \mathsf{A}, \mathsf{B}\}$, and (B1)–(B4) become unit and binary clauses.

Proposition 6.6. *No five-state rule that satisfies (B1)–(B4) in the weak reading, with* $I(\mathsf{F}) = \mathsf{F}$*, synchronizes all lines of lengths* $2, \dots, 12$ *in minimal time; a fortiori, no rule in Balzer's form does. The lengths up to* 11 *do not suffice to prove this.*

Proof. For each involution, the clauses of (B1)–(B4) are added to $\mathrm{MT}_5(\{2, \dots, N\}; 2N - 2)$. For $I = (\mathsf{A}\,\mathsf{B})$ and $N = 10$, and for $I = (\mathsf{L}\,\mathsf{A})$ and $I = (\mathsf{L}\,\mathsf{B})$ and $N = 9$, CaDiCaL refutes the formula with an LRAT proof. For I the identity and $N = 12$, Kissat refutes it; `drat-trim` turns its DRAT proof into an LRAT proof. Each of the four proofs is verified by `lrat-check`. The rule `results/balzer_sym11.txt` satisfies (B1)–(B4) in the strong reading with I the identity and synchronizes every line of length at most 11 (direct simulation). □

7 DISCUSSION

The two barriers constrain the half-line from opposite sides, and the five-state rules of Section 6.3 show how they complement each other. The half-line of δ_{12} meets the speed-1/3 barrier exactly in simulations, but its periodic region at the left border is excluded by the left-border barrier (Corollary 6.5). The half-line of δ_{14} follows the elementary cellular automaton 22 and shows a nested pattern (Figure 5); it passes both barriers in the simulated ranges, and the rule fails only in the first rows of the reflected triangle of the line of length 15. All five rules of Proposition 6.3 fail in the first rows of their reflected triangles.

The early rows and the right ends of the reflected triangles are controlled by Corollary 5.3 and Lemma 5.5, but they do not exclude every rule. For the germ of δ_{14}^{d}, which has complexity 15, conditions (G1)–(G3) with $T = 10$ and all $n \ge 12$ have a model, and so do (G1)–(G3) with $T = 7$ and all $n \ge 9$ together with (G5) for $9 \le n \le 20$. In the second model, the lines of lengths 9, $11, \dots, 17$ and 19 have both end cells, but not all cells, in state F at time $2n - 2$, and no cell in state F before (both models are in the supplementary material). This suggests that excluding such rules needs conditions on the middle cells of the firing row, which are expensive

to encode: for $n = 15$, the dependency cone of the middle cell of the firing row contains 76 of the 120 cells of R_{15}.

We close with four questions.

(1) Does a five-state minimal-time solution exist?

(2) Is there a condition on the middle of the reflected triangles that is as economical as Lemma 5.5?

(3) Can the bound $(k-1)^{2s}\ell$ of Lemma 4.1 be made polynomial in s?

(4) Do Balzer's conditions exclude six states?

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Data availability The programs, rule tables, germ lists and logs behind every computational statement, two scripts that rerun them, the solver times, proof sizes and SHA-256 digests, and the program that draws the figures are in the supplementary material and at `https://github.com/giamkiakeng/proj1` (directory `fssp5/paper/supplementary`).

NOMENCLATURE

Symbol	Meaning	Where
$S,\ W,\ k$	the set of states, the working states $S \setminus \{\mathsf{F}\}$, and $k = \lvert S\rvert$	Section 2
$\mathsf{L},\ \mathsf{G},\ \mathsf{F}$	the quiescent state, the general and the firing state	Section 2
$\mathsf{A},\ \mathsf{B},\ \mathsf{C}$	auxiliary states	Section 2
$*$	the border	Section 2
δ	a rule, defined on the neighbourhoods (x, y, z)	Section 2
$C_n(t, i)$	the state of cell i at time t in the line of length n	Section 2
$C_\infty(t, i)$	the same in the half-line	Section 2
R_n	the reflected triangle of the line of length n	Section 2
$\tau,\ \kappa$	relative time $t-(n-1)$ and distance $n-i$ to the right end	Section 3
$R_n(\tau, \kappa)$	the state of C_n at $(n-1+\tau, n-\kappa)$	Section 3
$J_n(\kappa)$	the input word: the cells of the half-line on $t+i = 2n-3$ and $2n-2$	Section 3
Ψ	the map that computes R_n from J_n	Lemma 3.1
$\mathrm{cone}_n(\tau, \kappa)$	the input cells on which $R_n(\tau, \kappa)$ depends	Section 3
$\mathrm{PUMP}(n, n')$	the conclusion of the pumping lemma	Lemma 3.3
$\Phi_t(j)$	the cell of the half-line at depth j behind the front at time t	Section 3
$\theta(j)$	the time from which depth row j is periodic	Theorem 3.4
$v_c(i),\ u_c(y)$	the cells of chain c, in direct and in skewed coordinates	Section 4
$E(T)$	the first $T+1$ rows of a reflected triangle	Section 5
$V_t,\ G_D$	the first $D+1$ depth rows at time t, and the map that advances them	Lemma 5.2
$\mathcal{G}_k(T, N_1, C)$	the early-row conditions (G1)–(G5)	Definition 5.4
$\mathrm{MT}_k(\mathcal{N}; M, N')$	the formula for the lines in $\mathcal{N}$, the half-line up to $t+i = M$ and PUMP up to N'	Section 6.1
$c_M(\delta)$	the number of neighbourhoods besides $(\mathsf{L}, \mathsf{L}, \mathsf{L})$ on the half-line up to $t+i = M$	Section 6.4
$\delta_{12},\ \delta_{13},\ \delta_{14}$	five-state rules for the lengths up to 12, 13 and 14	Section 6.3

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